LIEHRMANN Arnaud
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EDUCATION

2023 **PhD** in computational biology & statistics – SDSV, Uni. Paris-Saclay, France
2020 **Master** in computational biology – GENIOMHE, Uni. Paris-Saclay, France
2018 **Bachelor** in computational biology, Uni. Paris-Saclay, France

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CURRENT AND PAST POSITIONS

I have consistently worked in an interdisciplinary context between computational biology & statistics, in close collaboration with biologists.

Since 2024 **Postdoctoral researcher** in computational biology & statistics, CQSB, Sorbonne Uni., France.

- *"Interspecies analysis of transcriptom: methodological developments"* supervised by Élodie Laine (bioinformatics, Sorbonne Uni.) and Hugues Richard (statistics, RKI)
- Collaboration I initiated: Paul Bastide (statistics, Uni. Paris Cité), Mélina Gallopin (statistics, Univ. Paris-Saclay) and Maëlle Daunesse (genomics, Institut Pasteur)

2020–2023 **PhD student** in computational biology & statistics, Uni. Paris-Saclay, France. Link

- *"Multiscale analysis of transcriptome: methodological and algorithmic developments"* supervised by Guillem Rigaill (statistics) and Benoît Castandet (molecular biology)
- My work was conducted between:
  (1) a Plant genomics lab (IPS2)
  (2) an applied mathematics lab (LaMME)

2018-2020 **Research intern**

- I completed three research internships, including six months in the Machine Learning team at Northern Arizona Uni. in Flagstaff, **USA**, where I worked on supervised segmentation models applied to genomics data (ChIP-Seq). Link

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BIBLIOMETRY (according to Google Scholar Link)

Total: 12 articles + 3 *in preparation* (~ 77 citations)
Started after my PhD defense: 5 articles + 3 *in preparation*

RESEARCH ACHIEVEMENTS

My research focuses on the development and application of statistical models, algorithms, and methods for the analysis and interpretation of high-throughput biological data.

Computational Statistics

- Statistical inference: changepoint detection (segmentation), penalized likelihood
- Algorithms: dynamic programming, functional pruning

Biostatistics/Bioinformatics for Molecular Biology

- Models: segmentation, generalized linear models, phylogenetic comparative methods
- Data: RNA-Seq, Ribo-Seq, ChIP-Seq, RIP-Seq
- Package development: R/C++/python

1 - A. Liehrmann, E. Delannoy, B. Castandet and G. Rigaill. 2023. Link

DiffSegR: an RNA-Seq data driven method for differential expression analysis using changepoint detection. **NAR Genomics and Bioinformatics**

- DiffSegR is the segmentation method I developed during my PhD. From RNA-Seq data alone, it recovers the full set of annotated transcriptomic regulations in the Arabidopsis chloroplast, something no competing method achieves (best at 80%), while finding new ones. Its genericity is demonstrated through its application to sequencing data [4, 5], and its conclusions hold at nuclear genome scale [2].

Showcasing my expertise in applying segmentation methods to sequencing data (& others)

2 - V. Brunaud, G. Rigaill, and A. Liehrmann. ***manuscript in prep.***

DiffSegR2: genome-scale data-driven differential expression analysis

- We show that the method DiffSegR remains robust at the scale of whole nuclear genomes.

3 - M. Abakarova, M. I. Freiberger, A. Liehrmann, M. Rera and É. Laine. 2026. Link

Proteome-wide prediction of the functional impact of missense variants with ProteoCast. **Nature Communications**

- I applied segmentation to predicted protein mutational landscapes to identify putative binding and regulatory sites in unstructured regions, outperforming competing methods.

4 - K. Baudry, S. Skiada, C. Burstein, V. Liveanu et al. (incl. A. Liehrmann). 2026. Link

The GT1 domain of RNase J ensures RNA quality control through dsRNA binding in Arabidopsis plastids. **Nucleic Acids Research**

- Using DiffSegR on RNA-Seq data from multiple RNase J mutants in *Arabidopsis*, I showed that mutants accumulate high levels of dsRNA in the chloroplast.

5 - H. C. Tran, V. Schmitt, S. Lama, C. Wang et al. (incl. A. Liehrmann). 2023. Link
An mTRAN-mRNA interaction mediates mitochondrial translation initiation in plants.
Science

- Using DiffSegR on RIP-Seq data, I showed that *Arabidopsis* mTRAN factors bind preferentially to 5′ regions of mitochondrial mRNAs.

6 - A. Liehrmann, G. Rigaill and T. D. Hocking. 2021. Link
Increased peak detection accuracy in over-dispersed ChIP-seq data with supervised segmentation models. **BMC Bioinformatics**

- I showed that supervised segmentation of ChIP-Seq data yields higher-quality histone modification peak tracks than bioinformatics heuristics.

Showcasing my expertise in fine-grained modeling of transcriptomic and proteomic data

7 - A. Liehrmann, L. Carrel Billiard, P. Bastide, M. Gallopin, H. Richard and É. Laine. ***preprint.*** Link
phyloDS: An RNA-Seq Driven Method for Differential Splicing Analysis Across Species.

- First method for inter-species differential splicing analysis from RNA-Seq data.

8 - M. Gallopin, M. Daunesse, O. Lespinet, A. Liehrmann, P. Bastide. ***preprint.*** Link
Interspecies Differential Gene Expression Analysis with Regularized Phylogenetic Linear Model.

- First method to correctly control the false discovery rate for inter-species differential expression analysis from RNA-Seq data.

9 - L. Carrel Billiard, A. Liehrmann, H. Richard and É. Laine. ***manuscript in prep.***
Charting the Evolution of Protein Splice Variations Across More Than One Billion Years.

- First framework integrating evolutionary splicing graphs with protein-scale datasets to reconstruct the evolutionary history of alternative splicing across species.

10 - M. Guilcher, A. Liehrmann, C. Seyman, T. Blein et al. 2021. Link
Full length transcriptome highlights the coordination of plastid transcript processing.
International Journal of Molecular Sciences

- First method to detect transcriptome-wide co-maturations between pairs of RNA processing events from long-read RNA-Seq data.

Showcasing my expertise in developing statistically grounded and fast segmentation methods

11 - A. Liehrmann, G. Romano, P. Fearnhead, G. Rigaill. ***Poster GDR IASIS & manuscript in prep.***
Papillon: pruning dynamic programming for changepoint detection by comparing means

- A pruning rule for exact changepoint detection, as simple to implement as PELT and almost as efficient as functional pruning, that only compares segment means; C with R and Python ports.

12 - A. Liehrmann and G. Rigaill. 2024. Link
Ms.FPOP: a fast exact segmentation algorithm with a multiscale penalty.
Journal of Computational and Graphical Statistics

- A global optimization algorithm for multi-scale penalties implemented in R/C++.

Showcasing my commitment to making protocols and results accessible to the community

13 - M. Abakarova, M. I. Freiberger, A. Liehrmann, M. Rera and É. Laine. 2026. Link
ProteoCast: a web server to predict, validate, and interpret missense variant effects. **Journal of Molecular Biology**

14 - E. Delannoy, A. Liehrmann and B. Castandet. 2024. Link
The Use of Nanopore Sequencing to Analyze the Chloroplast Transcriptome Part II: Bioinformatic Analyzes and Virtual RNA Blots. **Plastids: Methods and Protocols, Springer US**

15 - S. Skiada, A. Launay-Avon, A. Liehrmann, E. Delannoy et al. 2024. Link
The Use of Nanopore Sequencing to Analyze the Chloroplast Transcriptome Part I: Library Preparation. **Plastids: Methods and Protocols, Springer US**

Showcasing my commitment to disseminating my research

I have 10 abstracts accepted as talks or posters in national and international conferences and workshops (JOBIM, SMPGD, Gordon Research Conference, SFBBM, JdS-SFdS, Paris-Saclay change-point workshop, German Conference on Bioinformatics).

PEER RECOGNITION

Invited short-term researcher in international laboratories

Since 2025	Robert Koch Institute, Berlin, **Germany**	(1 month)
2024	Department of Mathematics and Statistics, Lancaster Uni., **United Kingdom**	(1 week)

Invited presentations at lab-seminars

2025	Young Researchers' Seminar, Robert Koch Institute, **Germany**	(for bioinformaticians)
2024	LBCPLM Seminar, IBPC, France	(for biologists)
2024	MAV Seminar, LPSM, France	(for statisticians)

Invited presentations at conferences and workshops

2026	JOBIM, Strasbourg, France	(for bioinformaticians)
2024	K-mers mini symposium, Dijon, France	(for bioinformaticians)
2022	StatScale ECR meeting, Brighton, **United Kingdom**	(for statisticians)
2022	GRC post-transcriptional gene regulation, Maine, **USA**	(for biologists)

Invited PhD lecture

Since 2024 — Introduction to changepoint detection, application to genomics (for bioinformaticians)
- GDR BIMMM Early-Career Researchers' School, France

2025 — Introduction to changepoint detection from first principles (for statisticians)
- PhD Program Data Science, Uni. La Sapienza, Roma, **Italy**

2021 Introduction to differential expression analysis (for biologists)

- Genopole Summer School, France

Reviewer for journals and conferences

Since 2021 Reviewer for ISMB, ECCB, and NAR Genomics and Bioinformatics. (>10)

OTHER CONTRIBUTION TO RESEARCH

Supervision of Master students

2 co-supervised Master's students are now PhD students

- Benjamin Vacus (6 months, 2022) Link
- Chloé Seyman (3 months, 2021) Link

1 co-supervised Master's student will start her PhD in early 2027

- Milica Urosevic (6 months, 2025)

Teaching

Since 2022 Detection of transcriptional variants (Total: 12 h)

- First-year Master's students in Biology, Uni. Paris-Saclay

2020 Introduction to statistics (Total: 27 h)

- Second-year Bachelor's students in Biology, Uni. Évry Val d'Essonne